

Systems and embedded developer working close to the metal — Rust tooling published on crates.io, firmware and communications stacks in C, and hardware taken from schematic through 4-layer PCB layout. Final-year Computer Science and Business Systems undergraduate looking for low-level software or embedded engineering work.

■ EXPERIENCE

08/2025 – 04/2026
Bengaluru, IN

Software Engineering Intern

SonyTech IT Company — Robotics & Hardware Engineering

- Designed and developed a robotics and electronics educational robot with onboard AI processing for classroom use.
- Engineered the communications and networking stack, supporting UART, SPI and I²C for real-time data exchange.
- Created a 4-layer PCB hat built around an RP2350B microcontroller, acting as the Raspberry Pi 5 control interface.
- Executed schematic design, PCB layout and signal-integrity analysis against electromagnetic compatibility and performance standards; design completed and handed over to the team.

01/2023 – present
Belagavi, IN

Systems Programmer

Independent, open source

- Developed and published open-source tools and libraries in Rust and Python, including command-line utilities for text rendering, ASCII art and image processing.
- Built embedded and hardware projects end to end: binary input devices, IoT robots and a modular cyberdeck.

02/2025
Belagavi, IN

Web Development Lead & Design Co-Lead

Yukti 2k25 Techno-Cultural Fest, VTU Belagavi

- Led full-stack development of yukti.bastamasta.dev using HTML, CSS, JavaScript, GitHub Pages and Formspree.
- Managed content updates, the registration system and concert ticket sales through high-traffic periods.
- Co-led design, producing branding and marketing material for the event.

11/2023 – 01/2024
Belagavi, IN

Lead Student Designer

Yukti 2K24 National Level Techno-Cultural Fest, VTU Belagavi

- Led the student design team, managing workflow and aligning design goals with event organisers.
- Delivered posters, brochures and backdrops, holding consistent branding across all event material.

10/2020 – 07/2022
Remote

Freelance Developer

Fiverr

- Designed and delivered custom-modelled Discord bots for multiple clients, automating role assignment, moderation, raffles and giveaways.

■ SELECTED PROJECTS

Rust
since 04/2025

boxy-cli owner & lead dev 3.8K DOWNLOADS

A Rust crate for formatted command-line output — boxes with independent sections and divisions. Published on crates.io since April 2025, with around 3.8k downloads.

Embedded

BinaryKeeb

An embedded systems project: a display and two mechanical keys that send ASCII-encoded output to the connected system from binary input.

Rust

Brainfuck Interpreter · Image Processor CLI

An interpreter for the Brainfuck language, and a command-line image tool for cropping, inverting, grayscaling, rotating and brightening images. Both built in Rust.

Web

Portfolio Website

Personal site listing my work and projects, live at bastamasta.dev — links to every project here.

Python

Google Forms to Discord

Discord bots that manage Google Forms responses — polling for votes, notifying applicants of outcomes and related actions.

■ IN PROGRESS

since 02/2026
Hardware

ShellDeck

A custom cyberdeck with a swappable architecture for rapid system swaps: interchangeable hardware modules to streamline upgrades and maintenance, components chosen for portability, and firmware supporting seamless transitions between operating systems and configurations.

Python

boxy-py

A port of boxy-cli to Python — a library for building and displaying colourful text boxes in command-line interfaces.

IoT / AI

TARS Replica

An IoT and AI robot mimicking TARS from Interstellar, with voice recognition and commands, and AI-generated replies and actions.

Bash 5.2

Bash Web Server

A web server written entirely with Bash builtins, using no external utilities, serving local directory files and specific webpages for access to my system from anywhere.

Flutter · GML

MemeSoundPad · Action RPG

A multi-platform Flutter app of popular meme sound effects, and an adventure role-playing game built in GameMaker Studio 2 with GML.

■ SKILLS

Languages

Rust · C · C++ · Python · Assembly (8086) · Bash · PowerShell 7 · GML

Systems

Embedded systems & IoT · Schematic capture & PCB layout · UART / SPI / I²C · RP2350B · Raspberry Pi · OS development · Linux · NixOS · WSL

Software

Flask · Flutter / Android · Cargo · Docker · Komodo · Google Cloud · Firebase Studio · cron · Server & network administration

Tooling

Git · GitHub · GitHub Actions · GitHub Codespaces · GitLab · Gitea (self-hosted)

Design

UI design · Web design · Graphic design · Blender · Canva · Video editing

Spoken

English (professional) · Hindi (native)

■ EDUCATION

12/2022 – 09/2026

Belagavi, IN

B.Tech, Computer Science and Business Systems

Visvesvaraya Technological University — CGPA 8.94 / 10 (89.4%)

2022

Mangaluru, IN

Pre-University (PUC)

Expert PU College — 98%

2020

Kalaburagi, IN

High School, CBSE

Appa Public School — 86.4%

■ ACHIEVEMENTS & COMMUNITY

- Cloud Lead, Google Developer Groups, VTU Belagavi — guiding discussions and projects that build cloud computing knowledge in the community.
- First place, Code Sprint, IEEE VTU Chapter, 2023.
- District-level prize, TechRadiance 2020, for facial detection built with Python and OpenCV.
- Core organiser of Code Combat at Yukti 2k24, and event coordinator for Debugging at Yukti 2k25.